ELEC60012 – Electrical Energy Systems

Formula sheet

Part A

1. Coincidence coefficient for *n* households

$$j_n = j_\infty + \frac{1 - j_\infty}{\sqrt{n}}$$
$$P_n = j_n \times n \times P_1$$

2. Gauss-Seidel power flow update law

$$PQ\ bus: V_i^k = \frac{1}{Y_{ii}}\left(\frac{S_i^*}{(V_i^{k-1})^*} - \sum_{j=1}^{i-1} Y_{ij}\, V_j^k - \sum_{j=i+1}^{Nbus} Y_{ij}\, V_j^{(k-1)}\right)$$

Part B

1. Short circuit level in MVA

$$SCL\ (\text{in MVA}) = \frac{SCL_{base}\ (\text{in MVA})}{X_{eq}\ (\text{p.u.})}$$

2. Transformation matrix for sequence domain (a-phase) to phase domain and vice versa

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha^2 & \alpha \\ 1 & \alpha & \alpha^2 \end{bmatrix} \begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix}$$

$$\begin{bmatrix} V_{a0} \\ V_{a1} \\ V_{a2} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

The same transformation matrices are valid for current as well

3. Swing equation

$$\left(\frac{2H}{\omega_s}\right)\frac{d^2\delta}{dt^2} = P_m - P_e\ \text{p.u.}$$